

1. “If a CSP is k -consistent, then it must be $(k - 1)$ -consistent.” - Do you agree with the statement? If you do, provide a brief justification. If you do not, provide a counterexample with a brief explanation. 1 + 4
(CO1)
(PO1)

Solution:

The statement is False.

Consider the following CSP formulation:

Variables: X, Y

Values: $X = \{Red, Green, Blue\}$ and $Y = \{Red, Green, Blue\}$

Constraints: $X \neq Y$ and $X \neq Blue$

The formulated CSP is not 1-consistent since the domain of the variable X contains the color Blue. However, it is 2-consistent since for each color in the domain of X , we can find a color in the domain of Y assigning which does not violate the constraint involving the pair of variables.

Rubrics:

- 1 point for the correct decision
- 2 points for the problem formulation
- 2 points for the brief explanation

2. Assume that we have a tree-structured CSP with 10 variables. Each variable can take 50 values. You are running your CSP solver on a computer that can process 10^9 nodes/second. Determine the time required to solve the CSP using following algorithms: 4
(CO1)
(PO1)

a) Naive Backtracking

Solution:

$$\begin{aligned}\text{Time Required} &= \frac{d^n}{10^9} \\ &= \frac{50^{10}}{10^9} \text{ seconds} \\ &\approx 3 \text{ years}\end{aligned}$$

Rubric:

- 2 points for the correct equation

b) Tree-CSP-Solver

Solution:

$$\begin{aligned}\text{Time Required} &= \frac{nd^2}{10^9} \\ &= \frac{10 \times 50^2}{10^9} \\ &= 25 \text{ microseconds}\end{aligned}$$

Rubric:

- 2 points for the correct equation

3. The 8-queens problem can be solved using the Hill-Climbing Search Algorithm. A single run of the algorithm requires t seconds, where t is the last 3 digits of your student ID. However, the algorithm is prone to getting stuck in a local maxima, ridges, or plateaus. In each case, the algorithm reaches a point at which no progress is being made.

a) Recommend a solution to avoid the problem.

4
(CO3)
(PO3)

Solution:

One possible solution is to use random-restart hill climbing. It conducts a series of hill-climbing searches from randomly generated initial states, until a goal is found. It is complete with probability 1, because it will eventually generate a goal state as the initial state.

Rubric:

- 2 points for proposal
- 2 points for explanation

b) Determine the expected time required to find the goal state considering your proposed solution.

2
(CO1)
(PO1)

Solution:

To determine the expected time required, we need to find the number of times we need to restart the algorithm until it finds the goal $\rightarrow$ Geometric Distribution. The expected number of restarts required is $\frac{1}{p}$, where p is the probability of success. Hence,

$$\begin{aligned}\text{Expected Time Required} &= t \times \frac{1}{p} \\ &= 401 \times \frac{1}{0.14} \text{ seconds} \\ &\approx 2864 \text{ seconds}\end{aligned}$$